Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

Ethyl isocyanide on hydrolysis in acidic medium generates

(A) methylamine salt and ethanoic acid

(B) ethylamine salt and methanoic acid

(C) propanoic acid and ammonium salt

(D) ethanoic acid and ammonium salt

Q.2

Which of the following statements is not correct?

(A) NF_3 is less basic than NH_3.

(B) Guanidine, $HN = C\left(NH_2\right)_2$, is a weak base.

(C) The bond angle in NH_3 is greater than in NF_3.

(D) The decreasing base strength of the amines $CH_3\left(CH_2\right)_2NH_2, CH_2 = CHCH_2NH_2$ and $HC \equiv CCH_2NH_2$ is $CH_3\left(CH_2\right)_2NH_2 > H_2C = CHCH_2NH_2 > HC \equiv CCH_2NH_2$

Q.3

Which of the following orders is true regarding the basic nature of NH_2 group?

(A) *o*-Toluidine > Aniline > *o*-Nitroaniline

(B) *o*-Toluidine < Aniline > *o*-Nitroaniline

(C) *o*-Toluidine < Aniline < *o*-Nitroaniline

(D) *o*-Toluidine > Aniline < *o*-Nitroaniline

Q.4

The secondary amine on treating with $NaNO_2$ and HCl produces

(A) diazonium salt

(B) nitrosamine

(C) nitrite salt

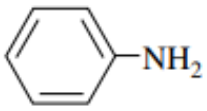
(D) isonitrite salt

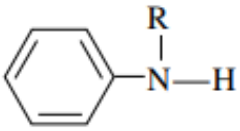
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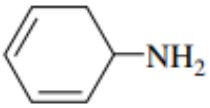
Q.5

Which of the following amines form N-nitroso derivative when treated with $NaNO_2$ and HCl.

(A) CH_3NH_2

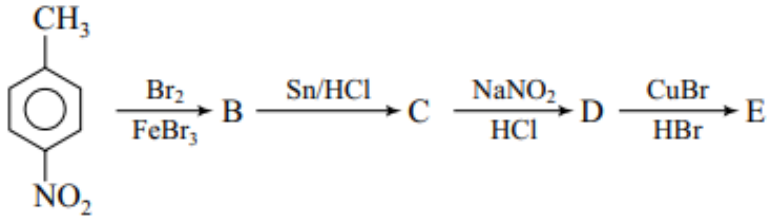
(B) 

(C) 

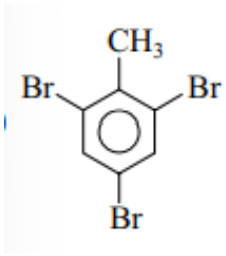
(D) 

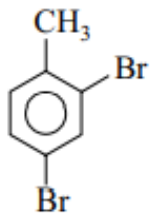
Q.6

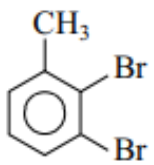
In a set of reactions p-Nitrotoluene yielded a product E



The product E would be

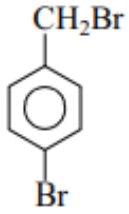
(A) 

(B) 

(C) 

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(D)



Q.7

On adding KI to benzenediazonium chloride, the product obtained is

(A) benzene

(B) 1,4-diiodobenzene

(C) iodobenzene

(D) $1,3,5$-triiodobenzene

Q.8

Which of the following statements is not correct?

(A) The main product obtained in the reaction of chloroform with aniline in the presence of excess of alkali is benzeneisonitrile.

(B) Amines are more basic than alcohols, ethers and esters.

(C) Nitrogen involves sp^2 orbitals in bonding with other atoms or groups in amines.

(D) Amines are less basic than water and also less basic than hydroxide ion.

Q.9

The correct order of basicity of the given compounds is

(A) $CH_3CH_2NH_2 < HO(CH_2)_3NH_2 < HO(CH_2)_2NH_2$

(B) $CH_3CH_2NH_2 < HO(CH_2)_3NH_2 > HO(CH_2)_2NH_2$

(C) $CH_3CH_2NH_2 > HO(CH_2)_3NH_2 > HO(CH_2)_2NH_2$

(D) $CH_3CH_2NH_2 > HO(CH_2)_3NH_2 < HO(CH_2)_2NH_2$

Q.10

Aniline is identified by

(A) Kolbe's reaction

(B) Riemer-Tiemann reaction

(C) Carbylamine reaction

(D) Haloform reaction

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Q.11

Which of the following statements is not correct?

(A) Primary amines show intermolecular hydrogen bonding.

(B) Secondary amines show intermolecular hydrogen bonding.

(C) Tertiary amines show intermolecular hydrogen bonding.

(D) Amines have lower boiling points as compared to those of alcohols and carboxylic acids of comparable molar masses.

Q.12

A positive carbylamine test is given by

(A) N, N-dimethylaniline

(B) 2,4 -dimethylaniline

(C) N-methyl- o-methylaniline

(D) p-methylbenzylamine

Q.13

Which of the following statements is correct?

(A) Hinsgberg test involves the reagent benzenesulphonyl chloride, in the presence of aqueous potassium hydroxide to distinguish primary, secondary and tertiary amines.

(B) Libermann reaction is shown only by primary amines.

(C) Primary amine gives liquid oxamide when treated with diethyloxalate.

(D) Secondary amine gives solid oxamide when treated with diethyloxalate.

Q.14

Which of the following statements is correct?

(A) Base-weakening substituents are the one that activate an aromatic ring toward electrophilic substitution.

(B) Aliphatic amine forms stable diazonium salt with nitrous acid.

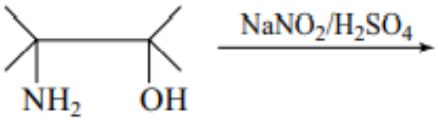
(C) All amino compounds can be distinguished by using the reagents chloroform and potassium hydroxide.

(D) The sulphonamide formed from secondary amine does not form sodium salt and is also insoluble in water.

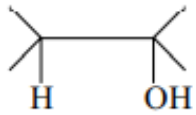
Q.15

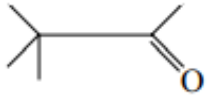
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The major product of the reaction

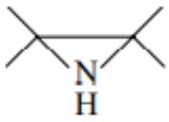


is:

(A) 

(B) 

(C) 

(D) 

Q.16

Which of the following can distinguish the three amines, viz., primary, secondary and tertiary?

(A) Azo-dye test

(B) Hinsberg reagent

(C) Carbylamine test

(D) Acetyl chloride

Q.17

Which of the following orders is correct regarding basicity of indicated molecules?

(A) N, N-Dimethyltoluidine $>$ p-toluidine $>$ aniline $>$ p-nitroaniline

(B) Aniline $>$ N, N-dimethyl- p-toluidine $>$ p-toluidine $>$ aniline

(C) p-Toluidine $>$ N, N-dimethyl $-p$-toluidine $>$ aniline $>$ p-nitroaniline

(D) N, N-Dimethyltoluidine $>$ aniline $>$ p-toluidine $>$ p-nitroaniline

Q.18

An organic compound having molecular mass 60amu is found to contain $20\%\text{C}, 6.67\%\text{H}$ and 46.67% N , while rest is oxygen. On heating it gives NH_3 alongwith a solid residue. The solid residue gives violet colour with alkaline copper sulphate solution. The compound is

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(A) $(NH_2)_2 CO$

(B) $CH_3CH_2CONH_2$

(C) CH_3NCO

(D) CH_3CONH_2

Q.19

Heating of $[CH_3CH_2CH_2CH(CH_3)N^+(CH_3)_3]OH^-$ produces

(A) $CH_2 = CH_2$.

(B) $CH_3CH_2CH_2CH = CH_2$

(C) $CH_3CH_2CH = CHCH_3$

(D) $CH_2 = CHCH_2CH = CH_2$

Q.20

The compound which on reaction with aqueous nitrous acid at low temperature produces an oily nitrosoamine is

(A) methylamine

(B) ethylamine

(C) diethylamine

(D) triethylamine

Q.21

The number of resonating structures of arylammonium ion is

(A) 2

(B) 3

(C) 4

(D) 5

Q.22

A compound with molecular mass $180u$ is acylated with CH_3COCl to get a compound with molecular mass $390u$. The number of amino groups present per molecule of the former compound is

(A) 2

(B) 5

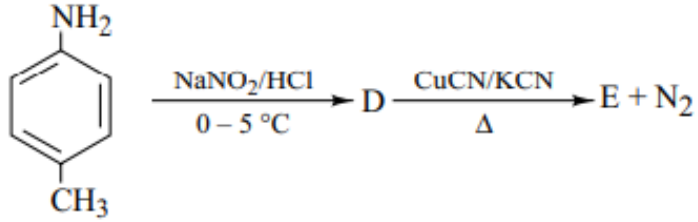
(C) 4

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(D) 6

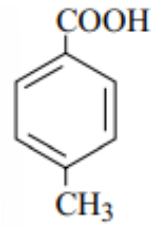
Q.23

In the reaction

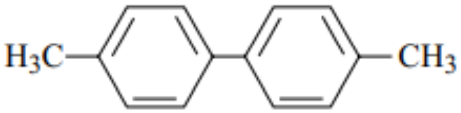


The product E is:

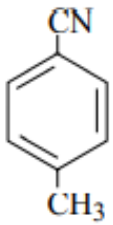
(A)



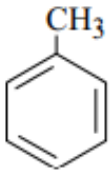
(B)



(C)



(D)



Q.24

In the Hofmann bromamide degradation reaction, the number of moles of $NaOH$ and Br_2 used per mole of amine produced are

(A) Four moles of $NaOH$ and two moles of Br_2

(B) Two moles of $NaOH$ and two moles of Br_2

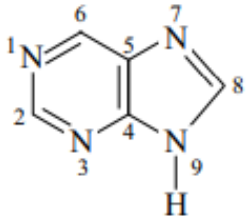
(C) Four moles of $NaOH$ and one mole of Br_2

(D) One moles of $NaOH$ and one mole of Br_2

Q.25

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The " N " which does not contribute to the basicity for the compound is.



(A) $\mathrm{N9}$

(B) $\mathrm{N3}$

(C) $\mathrm{N1}$

(D) $\mathrm{N7}$

Answers & Solutions

Q.1 Answer:

ethylamine salt and methanoic acid

Solution:

$$C_2H_5NC \xrightarrow[H^+]{H_2O} C_2H_5NH_2 + HCOOH$$

Q.2 Answer:

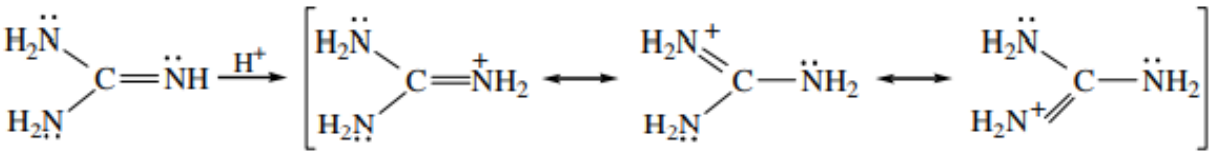
Guanidine, $\mathrm{HN}=\mathrm{C}\left(\mathrm{NH}_2\right)_2$, is a weak base.

Solution:

Explanation of options

(a) In NF_3, nitrogen acquires δ^+ charge because it is less electronegative than fluorine. With positive charge on nitrogen, the lone pair of electrons are more tightly held in NF_3.

(b) In guanidine, the imino $(=\mathrm{NH})$ nitrogen is sp^2 hybridised while amino $\left(-\mathrm{NH}_2\right)$ is sp ^{3} hybridised. The proton is likely to be attached to $-\mathrm{NH}_2$ as its nitrogen has less s-character. Actually the imino nitrogen is protonated because this leads to the stable symmetrical resonating cation with very high delocalisation energy as there are three contributing structures.



(c) Ammonia involves more bonding pair-bonding pair electronic repulsion as the pairs are located near the nitrogen atom due to its more electronegative character. In NF_3, F is more electronegative and thus bonding pair lies more near F leading to less electronic repulsion.

(d) The larger the s-character of a hybrid orbital of carbon, the larger electron-withdrawing (by induction) effect it has, consequently, the larger is its base-weakening effect.

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Q.3 Answer:

o-Toluidine < Aniline > *o*-Nitroaniline

Solution:

Electron-releasing group increases the availability of lone pair of electrons on nitrogen while electron attracting group decreases this availability.

The correct order of basic strength of aromatic amines is o − Toluidine < Aniline > 0− Nitroaniline.

Aniline is more basic than ortho nitro aniline because the presence of electron withdrawing nitro group decreases the ease with which N can donate a lonepair of electrons.

o-Toluidine is less basic than aniline because whenever a substituent is present in the ortho position, it decreases its basicity.

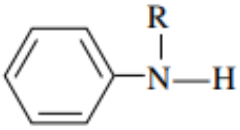
Q.4 Answer:

nitrosamine

Solution:

A nitrosamine is produced.

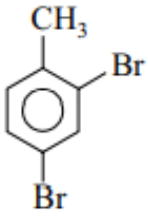
Q.5 Answer:



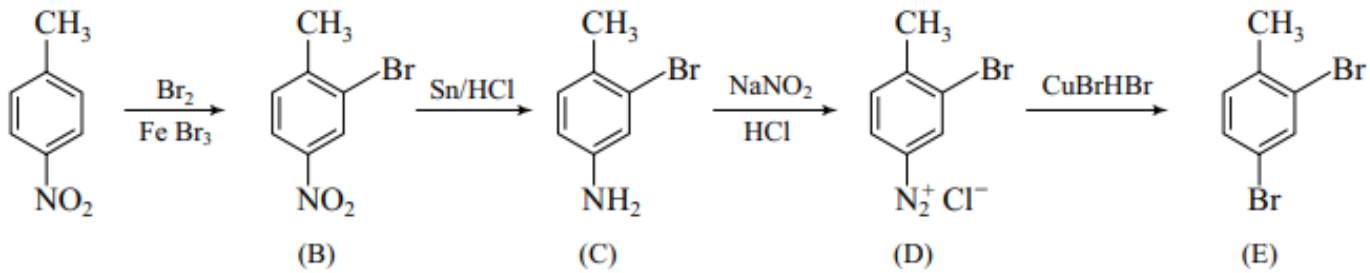
Solution:

2°Amine produces N-nitroso derivative.

Q.6 Answer:



Solution:



Q.7 Answer:

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iodobenzene

Solution:

When benzene diazonium chloride reacts with potassium iodide, iodobenzene is obtained. Diazonium group is replaced with $-\mathrm{I}$.

$$\mathrm{C_6H_5-N_2^+Cl^- + KI \xrightarrow{\Delta} C_6H_5-I + N_2\uparrow + KCl}$$

Q.8 Answer:

Amines are less basic than water and also less basic than hydroxide ion.

Solution:

Amines are more basic than water.

Q.9 Answer:

$\mathrm{CH_3CH_2NH_2 > HO\left(CH_2\right)_3NH_2 > HO\left(CH_2\right)_2NH_2}$

Solution:

The $-$ OH group causes decrease in the basicity of $-\mathrm{NH_2}$ group.

Q.10 Answer:

Carbylamine reaction

Solution:

Aniline is identified by carbylamine reaction. $\mathrm{RNH_2 + CHCl_3 + 3KOH \longrightarrow R-NC + 3KCl + 3H_2O}$.

Q.11 Answer:

Tertiary amines show intermolecular hydrogen bonding.

Solution:

Tertiary amine does not contain H attached to N.

For hydrogen bonding to occur, we need a polarized hydrogen atom (a hydrogen atom attached directly to F, O or N is called as a polarized hydrogen).

In a tertiary amine, there is no polarized hydrogen because all the hydrogens are replaced by alkyl or aryl groups.

So, tertiary amines cannot show hydrogen bonding.

Q.12 Answer:

p-methylbenzylamine

Solution:

Primary amine exhibits carbylamine test.

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Q.13 Answer:

Hinsgberg test involves the reagent benzenesulphonyl chloride, in the presence of aqueous potassium hydroxide to distinguish primary, secondary and tertiary amines.

Solution:

Explanation of options:

(b) Secondary amines show Libermann reaction.

(c) Solid oxamide is obtained.

(d) Liquid oxamide is obtained. So, the only correct option is (a)

Q.14 Answer:

The sulphonamide formed from secondary amine does not form sodium salt and is also insoluble in water.

Solution:

Explanation of options

(a) Electron-releasing groups are aromatic-ring activators and they also push electrons toward nitrogen making lone pair of electrons more readily available for an acid. Hence, they are base-strengthening groups. Electronwithdrawing groups act in the opposite direction, hence, they are base-weakening group.

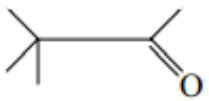
(b) Aliphatic amines form unstable diazonium salt which decompose to give nitrogen and aliphatic alcohols.

(c) Only primary amine can be distinguished as it produces carbylamine having most offensive smell

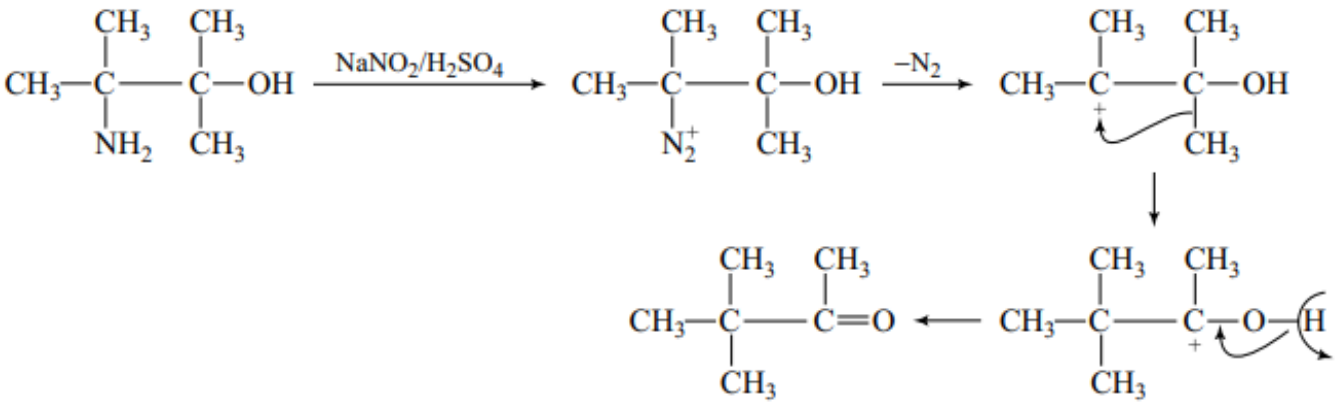
$$RNH_2 + CHCl_3 + 3KOH \rightarrow RNC + 3KCl + 3H_2O$$

So, the only correct option is (d)

Q.15 Answer:



Solution:



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Q.16 Answer:

Hinsberg reagent

Solution:

Hinsberg reagent can distinguish the three amines.

Q.17 Answer:

N, N-Dimethyltoluidine $>$ p-toluidine $>$ aniline $>$ p-nitroaniline

Solution:

Aniline is a weaker base. It is a Lewis base, i.e. the lone pair present on nitrogen can be given to the Lewis acids. The electron-releasing group present at the para position increases the electron density on nitrogen and thus makes it more basic. For example, para-toluidine is more basic than aniline. On the other hand, the presence of electron-withdrawing group decreases the electron density on nitrogen and hence makes it less basic. For example, p-nitroaniline is less basic than aniline.

Q.18 Answer:

$(NH_2)_2\,CO$

Solution:

The ratio of atoms is

$$C : H : N : O :: \frac{20}{12} : \frac{6.67}{1} : \frac{46.67}{14} : \frac{26.26}{16} :: 1.67 : 6.67 : 3.19 : 1.64 :: 1 : 4 : 2 : 1$$

Hence, empirical formula is $CH_4\,N_2O$

The reactions shown by the compound on heating is that of urea.

$$H_2NCONH_2 \xrightarrow[\Delta]{} H-N=C=O + NH_3$$
$$H_2NCONH_2 + H-N=C=O \rightarrow H_2NCONHCONH_2$$

Q.19 Answer:

$CH_3CH_2CH_2CH=CH_2$

Solution:

In Hofmann elimination, a less substituted alkene is formed as a result of the loss of more acidic BH (which follows the trend $1^\circ > 2^\circ > 3^\circ$). The reaction is

$$CH_3CH_2CH_2CH\,(CH_3)\,N^+\,(CH_3)_3 \longrightarrow CH_3CH_2CH_2CH=CH_2$$

Q.20 Answer:

diethylamine

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Solution:

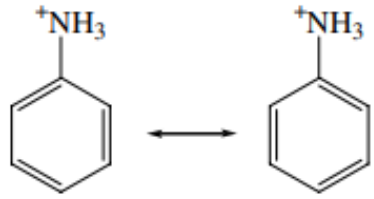
2° **amine produces oily nitrosoamine.**

Q.21 Answer:

2

Solution:

There are only two resonating structures.



Q.22 Answer:

5

Solution:

Let the compound be $\mathrm{R}\left(\mathrm{NH_2}\right)_x$**. Its acetylation will give** $\mathrm{R}\left(\mathrm{NHCOCH_3}\right)_x$**. Their molecular masses will be**

$$m_1\left(\mathrm{R}\left(\mathrm{NH_2}\right)_x\right) = m_{\mathrm{R}} + (14\mathrm{u})_x$$

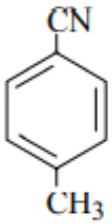
$$m_2\left(\mathrm{R}\left(\mathrm{CHCOCH_3}\right)_x\right) = m_{\mathrm{R}} + (56\mathrm{u})_x$$

Increase in molecular mass $= (56\mathrm{u} - 14\mathrm{u})_x = (42\mathrm{u})_x$

Given increase in the molecular mass $= 210\mathrm{u}$

Hence, $x = \frac{210\mathrm{u}}{42\mathrm{u}} = 5$

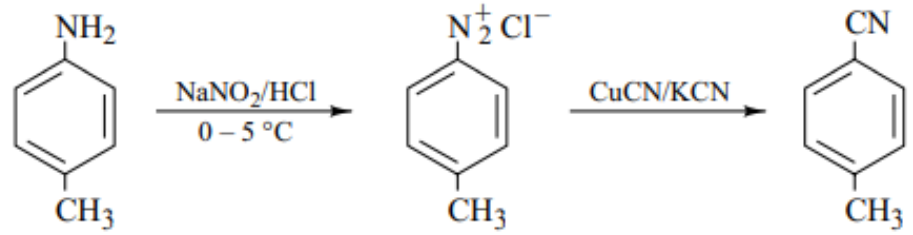
Q.23 Answer:



Solution:

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The reaction are:



Q.24 Answer:

Four moles of NaOH and one mole of $\mathrm{Br_2}$

Solution:

The net reactions is $\mathrm{RCONH_2 + Br_2 + 4NaOH \rightarrow RNH_2 + 2NaBr + Na_2CO_3 + 2H_2O}$

Q.25 Answer:

$\mathrm{N9}$

Solution:

The lone pair of electron on $\mathrm{N-9}$ **is involved in delocalization and does not contribute in the basicity for the compound.**

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